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Question 5 **(25 marks)**

The first five numbers in a pattern of numbers are given in the table below.

Term	Number
U_1	13
U_2	15
U_3	19
U_4	25
U_5	33
U_6	
U_7	
U_8	

(a) **(i)** Follow the pattern in the table above to write the next three numbers into the table.

(ii) Use the data in the table to show that the pattern is quadratic.

(b) $U_n = n^2 + bn + c$ where $b, c \in \mathbb{Z}$. Find the value of b and the value of c.

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(c) The table below shows the first five terms of an arithmetic sequence.
Find an expression for T_n, the nth term of the sequence.
Hence, or otherwise, find the value of T_{30}, the 30th term.

Term	Number
T_1	12
T_2	14
T_3	16
T_4	18
T_5	20

T_n

T_{30}

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